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Optimization of warehouse picking to maximize the picked orders considering practical aspects

Kimiya Rahmani Mokarrari [a], Taraneh Sowlati [a,*], Jeffrey English [b], Michael Starkey [b]

[a] Industrial Engineering Research Group, Department of Wood Science, University of British Columbia, 2931-2424 Main Mall, Vancouver, British Columbia V6T-1Z4, Canada
[b] Fujitsu Intelligence Technology Limited, Vancouver, British Columbia, Canada

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ABSTRACT

Majority of previous studies on optimization of warehouse order picking lack real-world application and overlook complexities such as varying items' weight, workload balance, and heterogeneous pickers. To address these issues, in this study, mathematical programming models are developed based on two approaches to assign orders to pickers and determine pickers' routes in a North American consumer goods warehouse. Both approaches maximize the number of picked orders. However, in the simultaneous modelling approach, decisions are made in one model, while in the sequential modelling approach, first routing decisions then order assignment decisions are made. The sequential modelling approach compared with the simultaneous one yields better results in a shorter time. For the considered warehouse, the results suggest that the warehouse may not be able to handle more than 700 orders per shift without expanding its workforce. Furthermore, adding one picker increases the number of picked orders by 2%. Additionally, balancing pickers' workload is more effective when there are fewer than 600 orders available for picking per shift, because with a higher number of orders, pickers remain busy throughout the entire shift. The proposed mathematical modelling approaches enable warehouse managers and staff to process more orders with available resources (equipment, time, staff) and make better decisions using available data.

1. Introduction

Warehouses and distribution centers play an important role in supply chains and logistics systems [1]. The primary operations within these facilities encompass receiving products, managing inventory, assigning storage location to products, picking, sorting, shipping and delivering orders [2]. In most warehouses, order picking, which refers to retrieving items from storage locations to fulfill customers' orders, is a key operation [2]. Widely recognized as one of the most labor-intensive, time-consuming, and costly processes in warehouses [3,4], order picking can account for more than 50% of total warehouse operating costs [5,6]. Therefore, optimizing the order picking process can potentially reduce costs, enhance efficiency in warehouses, and facilitate the timely delivery of products, which are essential for the survival of companies in today's competitive market [7].

In order to improve the order picking process, which can significantly impact warehouse efficiency [3,8], different optimization

* Corresponding author.
E-mail address: taraneh.sowlati@ubc.ca (T. Sowlati).

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models and heuristic methods have been developed in the literature. These models and methods were primary designed to make decisions regarding the assignment of orders to pickers, batching of orders, and routing of pickers to streamline the order picking process and reduce associated costs [3]. While research on these decisions contributes to the literature, findings of academic papers are often unacceptable to warehouse managers as academic papers often overlook the practical aspects of order picking [9]. Pinto and Nagano [4] also stated that the majority of the proposed models for order picking problems in the literature have not been applied to real case studies nor have considered the complexities of warehouses in real applications.

This paper is based on a real case study of a warehouse located in North America in which various complexities and practical aspects of real-world warehouses have been considered. These practical aspects include the presence of multiple pickers, variations in picking vehicle and product properties, and workload equity considerations, which are often overlooked in picking planning [4,9,10]. Workload balancing becomes a critical issue in warehouses where large quantities of orders are picked daily within a specific time limit [10]. Planning order picking operations and improving efficiency in warehouses inevitably raise concerns about workload imbalance [11]. While workload balancing has been a popular research topic within manufacturing or assembly lines, it has been less explored in the context of warehouse picking [10]. Accounting for these factors in picking models is crucial as overlooking them can lead to infeasible solutions in real-life cases [4,12].

In our case study, the goal of warehouse managers was to maximize the number of picked orders (i.e., fulfilled orders) in each working shift, while having a balanced workload for the pickers. It is worth noting that, in this study, the picker's workload is measured by the total working time of each picker in a shift.

To achieve the warehouse managers' goal, this study provides order picking planning models that determine the assignment of orders to pickers, the sequence of orders that should be picked, and the routing of pickers within the warehouse. To the best of authors' knowledge, previous models proposed for order picking have primarily focused on objectives including minimizing the picking time, tardiness, picking distance, or costs for a fixed number of orders. To minimize these objectives, it is essential to assume a fixed number of orders that should be fulfilled and assigned to pickers and to treat this quantity as an input to the models.

Therefore, existing studies fail to address the critical question of determining the number of orders a warehouse can handle and fulfill using its available resources, which is an essential piece of information for the managers. This gap highlights the absence of models capable of determining the maximum number of orders that can be fulfilled from the available customer orders within a specific shift, given a certain number of pickers. Maximizing the number of fulfilled orders can have an indirect effect on reducing the picking time for each order. This is because, in order to fulfill a greater quantity of orders, it becomes essential to pick orders more quickly. Having a clear understanding of the maximum number of orders that the warehouse can handle helps managers effectively plan the picking process, appropriately allocate resources, and mitigate the risk of unfulfilled orders.

To address the problem at hand, mathematical models using two approaches (simultaneous and sequential) are developed and tested on the case study. The performances of these models are evaluated and compared. These mathematical models will provide a novel decision-making tool designed to align with real-world scenarios, offering valuable insights for warehouse managers. This work makes significant contribution both to the industry and the existing order picking literature. Specifically, this study contributes to the industry by providing a decision-making support tool that helps managers make informed and improved decisions. Additionally, it contributes to the literature by considering a different objective function for picking planning and presenting two mathematical programming models that approach the same problem from different perspectives, emphasizing the importance of the effective mathematical modelling. Furthermore, the proposed models illustrate the relationship between pickers' workload balance, the number of pickers, and the number of orders that can be completely fulfilled within a working shift, which has not been addressed in the literature.

2. Literature review

The focus of this literature review is on picker-to-parts systems, considering that the majority of current picking systems still operate on a picker-to-parts basis, wherein pickers collect ordered items [4,9]. This prevalence is due to the high costs associated with transitioning to automated picking systems [13].

In addition to the picking system, different policies can be employed for the picking process. Batch picking is a policy where orders are first grouped into batches and then batches are assigned to pickers, allowing them to pick multiple orders simultaneously. Wave picking refers to a policy where orders with the same shipment due dates are picked together. In the zone picking, the warehouse is divided into different zones, then each picker is assigned to a specific zone to pick items from orders located in that zone. Lastly, strict order picking is a policy where pickers traverse the warehouse to pick the items of one order in each tour [14,15].

Majority of papers in the context of warehouse picking optimization focused on batch picking policy. In a batch picking policy, decisions such as order batching, batch assignment, batch sequencing, and picker's routing can be addressed to meet a certain criterion, such as time minimization [4]. Simultaneous consideration of these decisions leads to more complex mathematical programming models (e.g., [16–18]). Wave picking is often linked with batch picking, as the picking time of a wave can be significantly affected by how orders are batched [19]. Previous studies, such as [19] and [20] addressed order batching, batch assignment and picker's routing decisions, within a warehouse considering wave picking policy.

The zone picking policy can improve the travel time, congestions in the warehouse, and consequently the throughput of a picking system [21]. Various warehouse picking decisions have been investigated in the context of zone picking, including storage assignment [22], scheduling [10], routing [21], and batching [23]. Similar to wave picking, zone picking has often been addressed simultaneously with batch picking in each zone.

In a strict picking or picking-by-orders policy, the picking process is less prone to errors. However, it can be costlier and more time-

Table 1
Papers related to warehouse order picking grouped based on different picking policies. (References (6,10,15–23,26–40) is cited in Table body part).

Picking policy	Picking decisions	Number of pickers	Objective function	Practical factors: Workload equity	Practical factors: Varying picking vehicle property	Practical factors: Product properties	Case study	Reference
Batch picking	• Order batching • Batch assignment • Batch sequencing	Many	Min. tardiness					[32]
	• Order batching • Picker routing	One	Min. travelled distance					[33, 36]
		Many	Min. costs		Capacity			[34]
	• Order batching • Batch assignment • Picker routing	Many	Min. tardiness		Capacity	Order weights	✓	[6]
	• Order batching • Batch sequencing • Picker routing	One	Min. tardiness			Order weights		[35]
	• Order batching • Batch assignment • Batch sequencing • Picker routing	Many	Min. tardiness					[17]
			Min. picking time				✓	[18]
			• Min. travelled distance • Min. travelled time • Min. tardiness			Order capacities		[30]
			• Min. travelled distance • Min. tardiness • Min. travelled time, earliness and tardiness		Capacity	Order capacities		[31]
			Min. pickers' completion time			Product weights Product search & pick time		[16]
Wave picking /Batch picking	• Order batching • Batch assignment • Picker routing	Many	Min. makespan				✓	[19]
			• Min. makespan • Min. active pickers (workload)					[20]
	• Load assignment • Picker routing		Min. costs		Capacity			[37]
Zone picking	Routing	Many	Min. max. travelled time	✓				[21]
	Storage assignment		• Min. replenishment • Min. total workload imbalance	✓				[22]
	Scheduling	-	Min. workload imbalance	✓				[10]
	Order batching	-	• Min. picking time • Min. standard deviation of picking time	✓				[23]
Strict picking	Picker routing	One	Min. travelled distance					[27] [28] [15]
			Min. energy consumption			Product weights		[26]
		Many	Min. travelled distance			Product weights & volume		[38]
			Min. order service time					[39]
	• Storage assignment • Picker routing	One	Min. costs					[40]
		Many	Min. picking time		Travelling speed	Product weights		[29]

consuming compared to picking by batches [4]. Despite these disadvantages, the strict picking is still often preferred by firms due to its simple implementation and accuracy [24,25]. Moreover, for large order sizes, there is a little difference between the total picking time of strict picking policy and that of the batch picking policy [24]. This might be another reason why some firms prefer strict order picking policy over batch picking [24]. Despite the commonality and advantages of strict order picking, less attention has been given to this picking policy in the literature. The reviewed papers in the strict picking category mostly focused on picker's routing decisions [15, 26–28]. There are also papers focusing on more complex problems that include simultaneous storage assignment and picker routing decisions [29].

Regardless of the picking policy, incorporating multiple pickers increases the complexity of models. However, it can provide a more accurate representation of real-world warehouse scenarios. When multiple pickers are considered under order batching policy, the batch assignment decision is usually addressed in order to specify the pick list of each picker as observed in [6,16–18,30–32]. Majority of papers in batch picking, wave picking, and zone picking have considered multiple pickers. The strict order policy is the only category with more papers considering one picker.

As shown in Table 1, the majority of papers, regardless of the picking policy, focus on models aimed at minimizing time-related aspects of the picking process. This includes minimization of tardiness, travel time, make-span, picking completion time, and even travelled distance. A few other papers aimed at minimizing costs as their objective function, and only one paper considered energy usage as the objective function [26]. The common assumption in such models is that a certain number of orders should be picked within a specific time window, while minimizing the mentioned objectives (e.g., [6,19,20,31,33,34]). If this assumption is not met, no orders will be processed, as not picking any orders would result in the minimum travel time or distance (i.e., zero) and incur no picking costs.

The mentioned objective functions help design optimized picking plans that reduce the time and cost of order fulfillment. However, they fail to determine the maximum number of orders from the available customer orders (i.e., demand) that the warehouse can handle in each planning period with its available resources. It is crucial for warehouse managers to know this information to: (1) utilize the available resources (e.g., time, picking staff, etc.) appropriately, (2) avoid overloading the warehouse with too many orders, (3) guarantee timely customer order fulfillment, and (4) prevent costs associated with late delivery or unfulfilled orders. Moreover, maximizing the number of picked orders (i.e., fulfilled orders) eliminates the need for additional steps or tools to first indicate the number of orders that can be fulfilled and then minimize the picking time or cost for them. This aspect of order picking has not been investigated in the reviewed papers.

Another important conclusion from the literature review is that only a few papers considered multiple aspects of warehouse picking simultaneously (e.g., [20,22,23,31]). Accounting for multiple aspects of picking is important because satisfying one picking objective might be in conflict with other equally important ones. Therefore, there is a need to develop decision-making tools and multi-objective models that can cover different priorities and objectives of warehouse managers at the same time.

While the models in warehouse picking literature can potentially enhance the efficiency in a warehouse, the findings of academic papers are often not acceptable to warehouse managers as academic papers usually overlook the practical aspects of order picking [9]. Similar to the literature review conducted by Pinto and Nagano [4], our literature review concludes that majority of the proposed models for order picking problems in the literature have not been applied to real case studies nor have considered the complexities of warehouses in real applications.

The complexities of real warehouses can stem from varying picking equipment types, varying weights of items, and human factor challenges. Workload balancing is one of these human factor challenges in warehouses [10]. While workload imbalance can be an issue in warehouses with any picking policy, only papers related to zone picking policy have considered workload equity in their models. Another mentioned practical factor is considering pickers' properties, such as capacity of pickers or picking vehicles that can differ from one picker to another based on the equipment they use. This factor has been taken into account by a few studies, such as [29,31, 34], but often overlooked. Specific properties of items, such as weight or volume, is another important practical factor covered by some studies (e.g., [6,26]), but often overlooked in the literature. Some papers considered the weights of orders or the capacity required to fulfill each order (e.g., [6,30,31,35]), rather than focusing on the properties of each individual product. Accurate knowledge on such properties is essential, and incorporating them in planning models leads to the development of more practical models and greater usefulness of optimization results for warehouse managers. Varying characteristics such as capacity and travelling speed of picking equipment pieces have also been considered by a few studies (e.g., [6,29]).

In summary, this study addresses the concern of determining the maximum number of orders a warehouse can fulfill within a specific shift while balancing the workload for pickers. By considering picking aspects, such as multiple pickers with varying capacities, item weights, and workload balance, this research provides valuable insights for optimizing the picking process and resource allocation. Unlike existing studies that often overlook these picking aspects, the current research employs mathematical models to develop a decision-making tool aligned with real-world situations. The developed models in this study not only contribute to industry practices by offering managers informed decision-making support but also enrich the literature by introducing new perspectives on order picking optimization. Furthermore, the study sheds light on the relationship between pickers' workload balance, the number of pickers, and the fulfillment capacity of the warehouse, a dimension that has been largely unexplored in prior research.

Table 1 summarizes the reviewed papers based on picking policy, picking decisions, number of pickers, objective function, practical aspects, and whether or not a case study was considered.

3. Material and methods

3.1. Case study

This paper was inspired by a real case study of the warehouse of a consumer goods distribution company located in North America. The warehouse of this company is a multi-block warehouse with a rectangular shape and a total space of 9488 pallets. The warehouse has twelve main storage areas (i.e., blocks), each of which containing specific types of products. A schematic layout of the warehouse is provided in Fig. 1.

Orders received by the warehouse are picked by 12 pickers per shift (eight-hour). Pickers are divided into two groups based on the equipment they use. Eight pickers are pallet jack operators, and four pickers are forklift operators. A pallet jack has a capacity of 2500 kg and a forklift's capacity is double than that of a pallet jack. According to the data received from the warehouse, in the majority of picking cases, each order is picked by a single picker. Therefore, strict order picking policy is assumed as the picking policy for this warehouse.

Based on the daily order records, covering a period of eight months from January to August 2022, that the warehouse provided to us, the highest volume of orders received by the warehouse was 764 per day, and the lowest number of received daily orders was two orders, which were received during a weekend. The average daily number of orders that the warehouse receives is approximately 300 orders. Fig. 2 shows the daily orders received by the warehouse in eight months.

The warehouse management team were interested in a picking plan that maximized the volume of orders handled (i.e., fulfilled) on each shift with the same number of pickers. Moreover, they wanted to have a balanced workload for their pickers as much as possible and placed great emphasis on using all the pickers.

3.2. Problem definition

Based on the case study and communications with the warehouse managers, the optimization problem could be defined as a warehouse picking problem with a picker-to-part system and strict picking policy. The purpose is to maximize the volume of orders fulfilled (picked) on each working shift and to balance the workload of a fixed number of pickers while determining the assignment of orders to pickers and specifying the tours that pickers should take in the warehouse to collect the orders that were assigned to them. Determining the order assignment and pickers' routes also yields the working time of each picker in total.

The following conditions and assumptions are considered in the problem definition and mathematical modelling based on the current situation at the warehouse and the comments provided by warehouse managers:

- The warehouse has a dedicated storage strategy, where each item has a known and fixed storage location.
- Each picker has a limited load-carrying capacity, which varies based on their picking equipment.
- Pickers start their tours from a depot and return to it when picking an order is done.
- On each tour taken by a picker, one order is picked (i.e., strict order picking policy).
- Partial order picking is not allowed.
- There are enough items available in the warehouse to fill all orders on each shift.
- The searching time and retrieval time are considered in the travel time between two storage locations and are the same for all types of items. For example, the travel time between storage locations *i* and *j* includes the time that an average picker takes to go from *i* to *j* (x minutes) plus an additional y minutes for searching and retrieval at storage location *j*.
- Items in a specific storage location are the same.

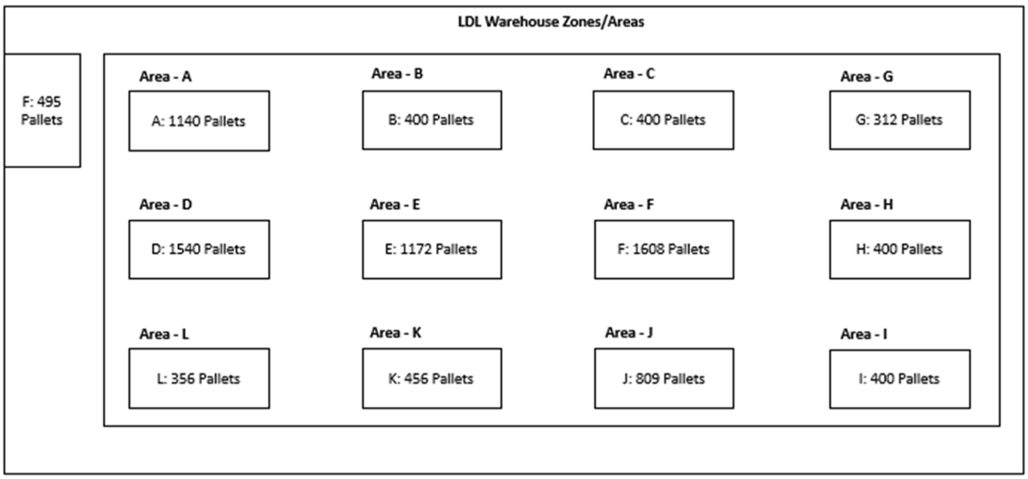

Fig. 1. Schematic layout of the warehouse.

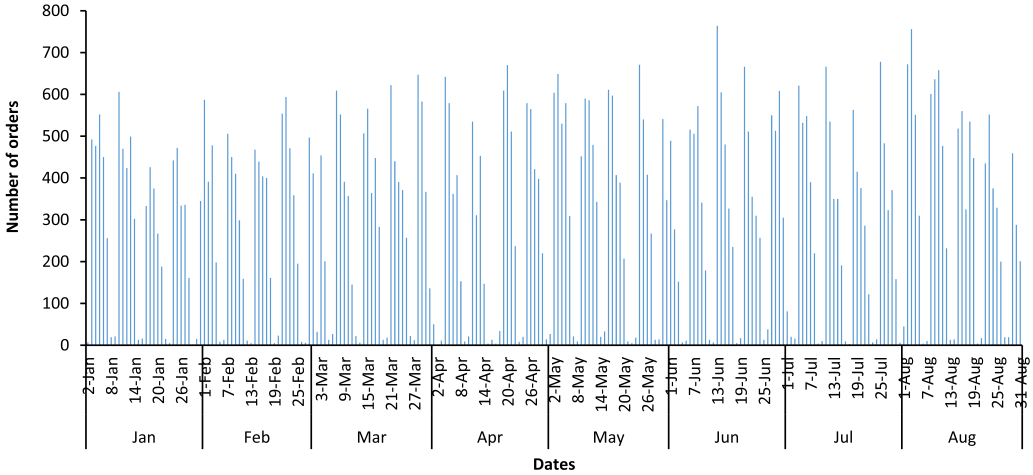

Fig. 2. Number of daily orders received by the warehouse from the beginning of January to the end of August 2022.

3.3. Mathematical models

In this section, two approaches are taken to provide mathematical models for the introduced problem. The simultaneous modelling approach makes order assignment and picker routing decisions at the same time to maximize the number of picked orders and minimize the imbalance workload among pickers. In the sequential modelling approach, first orders' routes are constructed with the objective of minimizing the routing time, and then orders are assigned to the pickers. The details of the two approaches and the rationale behind them are explained in following sections.

3.3.1. Simultaneous modelling approach

The elements of the developed simultaneous mathematical model, including the objective function, constraints and decisions are shown in Table 2.

The required sets, parameters, and variables to develop the model are listed in Table 3.

The first term in the objective function, which is shown in Eq. (1), maximizes the number of orders assigned to all pickers – in other words, fulfilled orders. The second term of the objective function balances the workload by minimizing the summation of differences between each picker's workload and the average workload, over all pickers. The same measure for workload balance was used in a log transportation optimization study [41] to balance the number of assigned truckloads to contractors. In order to estimate the average workload in our study, the following optimization model was run without considering the workload balance (i.e., excluding the second term of the objective function and constraint (20)), and then the obtained total picking time was divided by the number of pickers (i.e., $\frac{\sum_{o \in O} \sum_{r \in R} p_{or}}{|R|}$).

Table 2
Elements of the simultaneous modelling approach.

Simultaneous order assignment and routing
Objective function: • Maximize the assigned orders (i.e., fulfilled orders) to pickers and minimize workload imbalance Constraints: • Start and end the tours at depot • Visit storage location of all items of an order and visit each location once for each order • Tour continuity • Sub-tour elimination • Picker's capacity • Available picking time • Difference between pickers working time and the average workload Outputs (decisions): • Assignment of orders to pickers ($\boldsymbol{x_{or}}$) • Paths that each picker takes to collect items of an order ($\boldsymbol{y_{oijr}}$) • Completion time of picking an order by a picker ($\boldsymbol{p_{or}}$) • Total working time of each picker ($\sum_{o \in O} \boldsymbol{p_{or}}$)

Table 3
Sets, parameters, and variables used to develop the simultaneous mathematical model.

Sets	Description
$\boldsymbol{I}$	Set of storage locations (including the depot S) $\boldsymbol{i},\ \boldsymbol{j},\ \boldsymbol{k} \in \{1,2,\ldots,\boldsymbol{G}\} \cup \{\boldsymbol{S}\}$
$\boldsymbol{R}$	Set of pickers $\boldsymbol{r} \in \{1, 2, \ldots, \boldsymbol{R}'\}$
$\boldsymbol{O}$	Set of orders $\boldsymbol{o} \in \{1, 2\ldots, \boldsymbol{O}'\}$
Parameters	**Description**
$\boldsymbol{C_r}$	Load carrying capacity of picker $\boldsymbol{r}$
$\boldsymbol{D_{ij}}$	Travel time between storage locations $\boldsymbol{i}$ and $\boldsymbol{j}$
$\boldsymbol{Q_{oi}}$	Quantity of item from storage location $\boldsymbol{i}$ in order $\boldsymbol{o}$
$\boldsymbol{K_{oi}}$	1 if an item from storage location $\boldsymbol{i}$ is in order $\boldsymbol{o}$; 0 otherwise
$\boldsymbol{L_i}$	Load of an item from storage location $\boldsymbol{i}$
$\boldsymbol{M}$	Big M – a large number used in sub-tour elimination constraints
$\boldsymbol{TM}$	Maximum available time for picking
$\boldsymbol{AW}$	Average workload
Binary variables	**Description**
$\boldsymbol{x_{or}}$	1 if order $\boldsymbol{o}$ is assigned to picker $\boldsymbol{r}$; 0 otherwise
$\boldsymbol{y_{oijr}}$	1 if storage location $\boldsymbol{i}$ is visited right before storage location $\boldsymbol{j}$ in order $\boldsymbol{o}$ by picker $\boldsymbol{r}$; 0 otherwise
Positive variables	**Description**
$\boldsymbol{t_{oir}}$	Time at which picker $\boldsymbol{r}$ enters storage location $\boldsymbol{i}$ when picking order $\boldsymbol{o}$
$\boldsymbol{p_{or}}$	Picking completion time of order $\boldsymbol{o}$ by picker $\boldsymbol{r}$
$\boldsymbol{u_r}$	Absolute difference between the workload of picker $\boldsymbol{r}$ and the average workload

W_1 and W_2 are parameters and show the relative importance of the two terms in the objective function. The summation of W_1 and W_2 equals one.

$$Max.\ \left(W_1 \sum_{o\in O}\sum_{r\in R} x_{or} - W_2 \sum_{r\in R} u_r\right) \tag{1}$$

Constraint set (2) shows that each order can be assigned to at most one picker.

$$\sum_{r\in R} x_{or} \le 1\ \forall o \in \{1,\ldots,O'\} \tag{2}$$

Constraint set (3) accounts for the maximum load of items that each picker can carry when picking an order. In other words, when an order is assigned to a picker, the summation of the weights of all the items on that order needs to be less than or equal to the load-carrying capacity of the picker that has been assigned to that order.

$$\sum_{i\in G\cup S} Q_{oi}\ L_i\ x_{or} \le C_r\ \forall r \in \{1,\ldots,R'\}, o \in \{1,\ldots,O'\} \tag{3}$$

Constraint set (4) limits the number of times that each picker can visit a storage location to at most one for each order on each picking tour.

$$\sum_{i\in G\cup S,\ i\neq j} y_{oijr} \le 1\ \forall o \in \{1,\ldots,O'\}, j \in \{1,\ldots,G\} \cup \{S\}, r \in \{1,\ldots,R'\} \tag{4}$$

Constraint sets (5) and (6) show that if an order is assigned to a picker, the picker should start and end their picking tour at the depot.

$$\sum_{j\in G} y_{oijr} =\ x_{or}\ \forall i \in \{S\}, r \in \{1,\ldots,R'\}, o \in \{1,\ldots,O'\} \tag{5}$$

$$\sum_{i\in G} y_{oijr} =\ x_{or}\ \forall j \in \{S\}, r \in \{1,\ldots,R'\}, o \in \{1,\ldots,O'\} \tag{6}$$

Constraint set (7) indicates that if an order is assigned to a picker, meaning that order assignment variable (x_{or}) gets the value of one, then the picker should visit all the storage locations of the items that are on that order.

$$Q_{oj}\ x_{or} \le M\left(\sum_{i\in G\cup S} y_{oijr}\right)\ \forall o \in \{1,\ldots,O'\},\ j \in \{1,\ldots,G\}, r \in \{1,\ldots,R'\} \tag{7}$$

While constraint set (7) ensures pickers visit the storage location of items on the assigned orders (i.e., fulfilled orders), it does not necessarily avoid visiting storage location of items that are not on the assigned order's list. The reason is that when x_{or} is equal to zero,

y_{oijr} on the right-hand side can still get values. To avoid this situation, constraint set (8) needs to be added to ensure that pickers would only visit a storage location if that item is on the order's list assigned to them.

$$\sum_{i\in G\cup S} y_{oijr} \leq K_{oj} \ \forall o \in \{1,\ldots,O'\},\ j \in \{1,\ldots,G\}, r \in \{1,\ldots,R'\} \tag{8}$$

Constraint set (9) ensures the continuity of pickers' tours, which means that when a picker visits a storage location on a picking tour, they should leave that storage location to another one.

$$\sum_{i\in G\cup S,\ i\neq j} y_{oijr} = \sum_{k\in G\cup S,\ k\neq j} y_{ojkr} \ \forall o \in \{1,\ldots,O'\}, j \in \{1,\ldots,G\} \cup \{S\}, r \in \{1,\ldots,R'\} \tag{9}$$

Constraint set (10) prohibits tour creation for a single storage location (node).

$$y_{oijr} = 0 \ \forall o \in \{1,\ldots,O'\},\ r \in \{1,\ldots,R'\},\ i,j \in \{1,\ldots,G\} \cup \{S\}; i = j \tag{10}$$

Constraint sets (11) and (12) calculate the time at which a picker enters the first storage location after leaving the depot. When a picker goes from the depot to a storage location, t_{ojr} gets assigned a value after the first stop based on the traveling time between the first stop and the depot. It should be noted that constraint set (11) alone is not sufficient to indicate the time value (t_{ojr}) as it does not limit the value of t_{ojr}. Therefore, constraint set (12) is needed to create a limit for t_{ojr}, so that the two constraint sets make t_{ojr} equal to time at which a picker enters a storage location. M is a big number used in these constraints to neutralize the effect of these constraint in case $y_{oijr} = 0$.

$$t_{ojr} \geq D_{ij} - M\left(1 - y_{oijr}\right) \ \forall o \in \{1,\ldots,O'\},\ i \in \{S\}, j \in \{1,\ldots,G\}, r \in \{1,\ldots,R'\} \tag{11}$$

$$t_{ojr} \leq D_{ij} + M\left(1 - y_{oijr}\right) \ \forall o \in \{1,\ldots,O'\},\ i \in \{S\}, j \in \{1,\ldots,G\}, r \in \{1,\ldots,R'\} \tag{12}$$

Constraint sets (13) and (14) eliminate the subtours and calculate the time at which a picker enters a storage location while picking an order. The subtour elimination formulation used in this study is known as MTZ subtour elimination [42].

According to constraint set (13), between each consecutive pair of $i - j$ (i.e., storage locations, except for the depot), the time at which the picker enters storage location j should be higher than the time at which the picker visited storage location i. In this way, the unconnected subtours are prevented. Again, in here, constraint set (14) needs to be added to limit the value of t_{ojr} when $i - j$ is within the picking tour. These two constraint sets make the value of t_{ojr} equal to the time at which the picker visited previous storage location (t_{ojr}) plus the travel time between storage locations i and j.

$$t_{ojr} \geq t_{oir} + D_{ij} - M\left(1 - y_{oijr}\right) \ \forall o \in \{1,\ldots,O'\}, i,j \in \{1,\ldots,G\}, r \in \{1,\ldots,R'\} \tag{13}$$

$$t_{ojr} \leq t_{oir} + D_{ij} + M\left(1 - y_{oijr}\right) \ \forall o \in \{1,\ldots,O'\}, i,j \in \{1,\ldots,G\}, r \in \{1,\ldots,R'\} \tag{14}$$

Constraint sets (15) and (16) are similar to constraint sets (13) and (14). The only difference is that these constraint sets calculate the time at which the picker returns back to the depot.

$$t_{ojr} \geq t_{oir} + D_{ij} - M\left(1 - y_{oijr}\right) \ \forall o \in \{1,\ldots,O'\}, i \in \{1,\ldots,G\}, j \in \{S\}, r \in \{1,\ldots,R'\} \tag{15}$$

$$t_{ojr} \leq t_{oir} + D_{ij} + M\left(1 - y_{oijr}\right) \ \forall o \in \{1,\ldots,O'\}, i \in \{1,\ldots,G\}, j \in \{S\}, r \in \{1,\ldots,R'\} \tag{16}$$

While constraint sets (11) – (16) determine the times at which a picker visits a storage location (i.e., t_{ojr}) if that location is within pickers' tour (i.e., $y_{oijr} = 1$), they cannot limit the values of t_{ojr} to zero when a storage location is not visited in a picking tour (i.e., $y_{oijr} = 0$). Therefore, constraint set (17) is added to ensure that if a storage location is not visited by a picker, then the value of the picker's visiting time at that location is zero.

$$t_{ojr} \leq TM \sum_{i\in G\cup S} y_{oijr} \ \forall\ o \in \{1,\ldots,O'\}, j \in \{1,\ldots,G\} \cup \{S\}, r \in \{1,\ldots,R'\} \tag{17}$$

Constraint set (18) indicates the picking completion time of an order by a picker, which is equal to the time that a picker returns to the depot.

$$p_{or} = t_{oir} \ \forall o \in \{1,\ldots,O'\}, i \in \{S\}, r \in \{1,\ldots,R'\} \tag{18}$$

Constraint set (19) ensures that the completion time of all orders by a picker would not exceed the total available time for picking.

$$\sum_{o\in O} p_{or} \leq TM \ \forall r \in \{1,\ldots,R'\} \tag{19}$$

Constraint set (20) calculates the difference between a picker's workload (i.e., total picking time) and the average workload. With u_r in the objective function, this variable u_r will get the minimum possible value.

$$\left|\sum_{o\in O} p_{or} - AW\right| \leq u_r \ \forall r \in \{1, \dots, R'\} \tag{20}$$

As constraint set (20) makes the model non-linear, it is divided into the following two constraints shown in (20-a) and (20-b) to avoid non-linearity.

$$\sum_{o\in O} p_{or} - AW \leq u_r \ \forall r \in \{1, \dots, R'\} \tag{20-a}$$

$$\sum_{o\in O} p_{or} - AW \geq -u_r \ \forall r \in \{1, \dots, R'\} \tag{20-b}$$

The last two constraints, (21) and (22), show the domains and ranges of variables.

$$y_{oijr},\ x_{or} \in \{0,1\} \ \forall o \in \{1, \dots, O'\}, i,j \in \{1, \dots, G\}, r \in \{1, \dots, R'\} \tag{21}$$

$$t_{oir},\ p_{or}, u_r \geq 0 \ \forall o \in \{1, \dots, O'\}, i \in \{1, \dots, G\}, r \in \{1, \dots, R'\} \tag{22}$$

3.3.2. *Sequential modelling approach*

This approach is another way of looking at the problem at hand. As the picking policy of the warehouse of our case study is strict picking, the picking process of each order is independent of other orders. Therefore, instead of a single model that maximizes the number of picked orders, two separate models can be developed to serve the same purpose. That is, first, the picking route for picking each order is determined in order to minimize the picking time of each order individually. This model is executed for each order to determine the required time and tour for picking each of them. Then, in the second model, the orders (considering their required picking time) are assigned to pickers to maximize the number of assigned orders (i.e., fulfilled orders). In this way, instead of minimizing the picking time indirectly by maximizing the number of picked orders, the time aspect of picking is directly taken into account and minimized. Fig. 3 shows the steps of the sequential modelling approach.

The decision variables for the sequential model are shown in Table 4. The sets and parameters used in this model are the same as those defined in Table 3.

The mathematical model for each phase of the sequential modelling approaching approach is as follows:

Phase 1

The objective function (23) of this model is to minimize the travel time for picking. Constraint sets (24) – (36) serve the same purpose as constraint sets (5), (6), and (8) – (18) in Section 3.3.1.

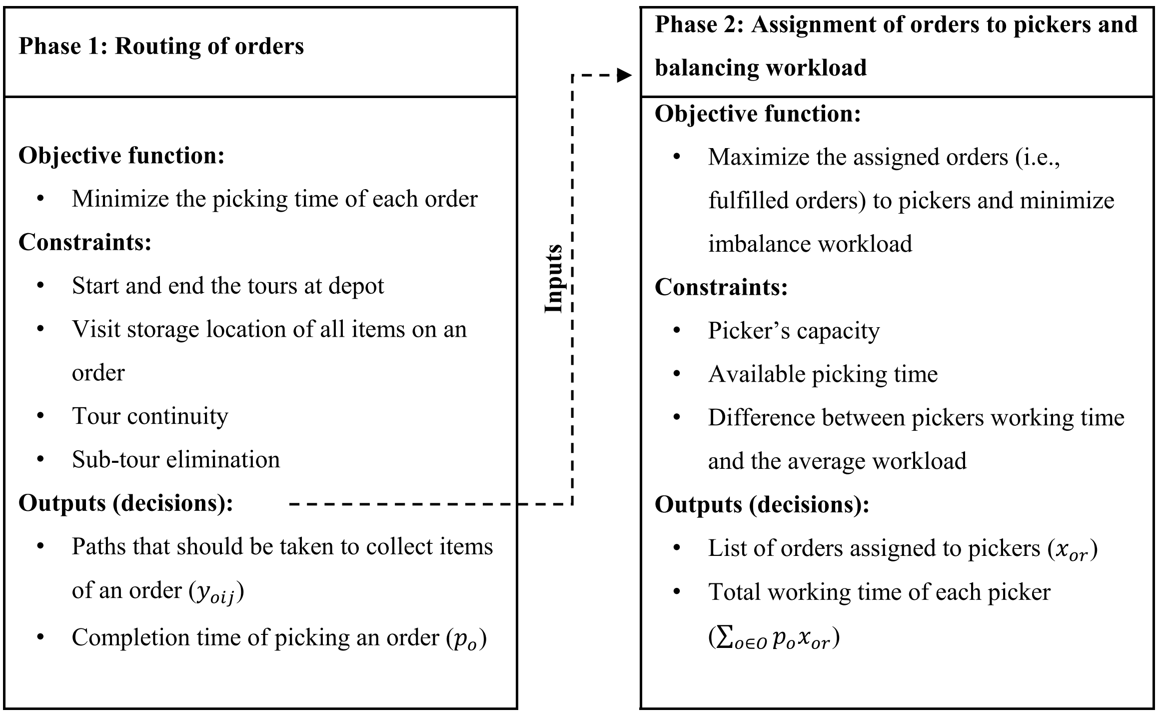

Fig. 3. Steps and elements of the sequential modelling approach.

Table 4
Variables used to develop the sequential mathematical model.

Binary variables	Description
x_{or}	1 if order o is assigned to picker r; 0 otherwise
y_{oij}	1 if storage location i in order o is visited right before storage location j in order o
Positive variables	**Description**
t_{oi}	Time at which storage location i is visited when picking order o
p_o	Picking completion time of order o
u_r	Maximum difference between the workload of picker r and the average workload

Note that the following model should be executed for each order (i.e., o) in the set of customer orders (i.e., {1, …, O′}):

$$Min. \sum_{i\in S\cup G}\sum_{j\in S\cup G} D_{ij}\, y_{oij} \tag{23}$$

Subject to:

$$\sum_{j\in G} y_{oij} = 1 \ \forall i \in \{S\} \tag{24}$$

$$\sum_{i\in G} y_{oij} = 1 \ \forall j \in \{S\} \tag{25}$$

$$\sum_{i\in G\cup S} y_{oij} = K_{oj} \ \forall j \in \{1,\dots,G\} \tag{26}$$

$$\sum_{i\in G\cup S} y_{oij} = \sum_{k\in G\cup S} y_{ojk} \ \forall j \in \{1,\dots,G\} \tag{27}$$

$$y_{oij} = 0 \ \forall i,j \in \{1,\dots,G\} \cup \{S\}; i = j \tag{28}$$

$$t_{oj} \ge D_{ij} - M\left(1 - y_{oij}\right) \ \forall\ i \in \{S\}, j \in \{1,\dots,G\} \tag{29}$$

$$t_{oj} \le D_{ij} + M\left(1 - y_{oij}\right) \ \forall\ i \in \{S\}, j \in \{1,\dots,G\} \tag{30}$$

$$t_{oj} \ge t_{oi} + D_{ij} - M\left(1 - y_{oij}\right) \ \forall i,j \in \{1,\dots,G\} \tag{31}$$

$$t_{oj} \le t_{oi} + D_{ij} + M\left(1 - y_{oij}\right) \ \forall i,j \in \{1,\dots,G\} \tag{32}$$

$$t_{oj} \ge t_{oi} + D_{ij} - M\left(1 - y_{oij}\right) \ \forall i \in \{1,\dots,G\}, j \in \{S\} \tag{33}$$

$$t_{oj} \le t_{oi} + D_{ij} + M\left(1 - y_{oij}\right) \ \forall i \in \{1,\dots,G\}, j \in \{S\} \tag{34}$$

$$t_{oj} \le TM \sum_{i\in G\cup S} y_{oij} \ \forall\ j \in \{1,\dots,G\} \cup \{S\} \tag{35}$$

$$p_o = t_{oi} \ \forall i \in \{S\} \tag{36}$$

Phase 2

The objective function (37) is the same as the model in Section 3.3.1. Constraint sets (38) – (41) serve the same purpose as constraint sets (2), (3), (19) and (20) in Section 3.3.1.

$$Max. \left(W_1 \sum_{o\in O}\sum_{r\in R} x_{or} - W_2 \sum_{r\in R} u_r\right) \tag{37}$$

Subjected to:

$$\sum_{r\in R} x_{or} \le 1 \ \forall o \in \{1,\dots,O'\} \tag{38}$$

$$\sum_{i\in G\cup S} Q_{oi}\, L_i\, x_{or} \le C_r \ \forall r \in \{1,\dots,R'\}, o \in \{1,\dots,O'\} \tag{39}$$

Table 5
Comparison between the results of the simultaneous and sequential modelling approaches.

Case scenarios	Case description	Number of received orders	Number of unique items in all the orders	Solution time (s)	Model results	
					Number of picked orders $\left(\sum_{o \in O} \sum_{r \in R} x_{or}\right)$	Total workload deviations (min) $\left(\sum_{r \in R} u_r\right)$
SM1	Without workload equity considerations (simultaneous modelling approach)	300 orders	183	470	298	799.47 [a]
SM1*	With workload equity considerations (simultaneous modelling approach)			1637	298	1.64
SQ2	Without workload equity considerations (sequential modelling approach)			480	300	2589.23 [a]
SQ2*	With workload equity considerations (sequential modelling approach)			480	299	21.27
SM2	Without workload equity considerations (simultaneous modelling approach)	500 orders	190	1506	496	327.47 [a]
SM2*	With workload equity considerations (simultaneous modelling approach)			1947	495	0.62
SQ2	Without workload equity considerations (sequential modelling approach)			720	500	2423.11 [a]
SQ2*	With workload equity considerations (sequential modelling approach)			720	499	11.06
SM3	Without workload equity considerations (simultaneous modelling approach)	700 orders	193	4004	668	14.80 [a]
SM3*	With workload equity considerations (simultaneous modelling approach)			Out of memory		
SQ3	Without workload equity considerations (sequential modelling approach)			1020	700	530.56 [a]
SQ3*	With workload equity considerations (sequential modelling approach)			1020	699	21.10

Table 5 footnote: SM1, SM2, and SM3 are case scenarios without workload balancing consideration under the simultaneous modelling approach. SM1*, SM2*, and SM3* are case scenarios with workload balancing consideration under the simultaneous modelling approach. SQ1, SQ2, and SQ3 are case scenarios without workload balancing consideration under the sequential modelling approach. SQ1*, SQ2*, and SQ3* are case scenarios with workload balancing consideration under the sequential modelling approach.

[a] This variable was not defined in the related mathematical model, and its value is calculated manually by determining the average workload in the respective case and calculating the total workload deviation from the average workload across all pickers.

$$\sum_{o\in O} p_o x_{or} \leq TM \ \forall r \in \{1,\ldots,R'\} \tag{40}$$

$$\left|\sum_{o\in O} p_o x_{or} - AW\right| \leq u_r \ \forall r \in \{1,\ldots,R'\} \tag{41}$$

Constraint set (41) should also be linearized similarly to the simultaneous modeling approach. Constraints (21) and (22) in Section 3.3.1 should also be added to show the domains and ranges of variables.

4. Results

In this section, the results of the proposed models are reported and compared. The models for both approaches are solved using a commercial optimization solver, Gurobi 9.5.2 (Copyright © 2022, Gurobi Optimization, LLC). The models were executed on a desktop computer with Intel ® Core (TM) i7–11,700 @ 2.50 GHz processor and 16.0 GB RAM.

4.1. Results of the simultaneous and sequential modelling approaches

Table 5 presents the results from both the simultaneous and sequential modelling approaches, comparing them across various scenarios, while each scenario is based on a specific number of orders received at the warehouse. In all scenarios, 12 pickers are considered, and the available picking time is an eight-hour shift, as detailed in the case study section.

It should be noted that in the execution and coding of the simultaneous modelling approach, decision variables y_{oijr}, which determines the routing of pickers for orders, are only defined for pairs of items i and j that exist in order o. This way, the number of required binary variables is reduced and the execution of the model is faster.

To show the importance of workload equity and to compare the order picking plans with and without workload equity considerations, each scenario is solved two times. SM1, SM2, and SM3 refer to the simulations modelling approach where only maximization

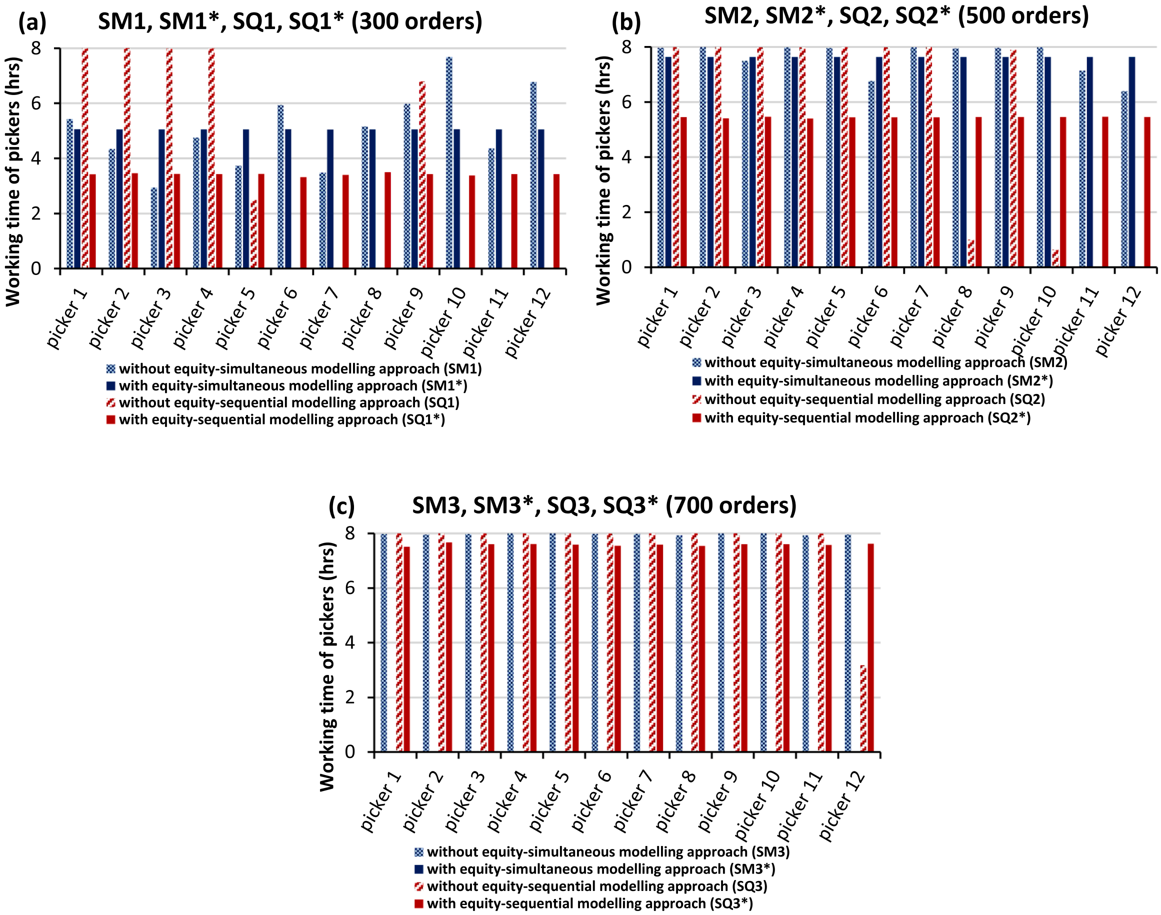

Fig. 4. (a-c). Comparison between the total working hours of pickers obtained from the simultaneous and sequential approaches.

of the picked orders is considered, which means that the objective function of the model is $Max. \sum_{o\in O}\sum_{r\in R} x_{o\ r}$, and SM1*, SM2*, and SM3* refer to scenarios where workload equity is also considered, which means that the objective function of the model is $Max. \left(W_1 \sum_{o\in O}\sum_{r\in R} x_{o\ r} - W_2 \sum_{r\in R} u_r \right)$. Similarly, SQ1, SQ2, and SQ3 refer to the sequential modelling approaching approach where only maximization of the picked orders is considered, and SQ1*, SQ2*, and SQ3* refer to scenarios where workload equity is also considered.

It should be noted that W_1 and W_2 are considered 0.9 and 0.1, respectively, because maximizing the number of picked (i.e., fulfilled) orders was the priority for the warehouse managers. Moreover, the average workload (AW) used in constraint set (20) of the model is calculated by dividing the total working time of all pickers obtained from solving the model without workload equity considerations by the number of pickers. The two terms of the objective function are also normalized to ensure they have the same range and effect in the objective function.

The simultaneous modelling approach requires a huge memory space, which led to an out-of-memory error when 700 orders were considered as input. This shows the limitation of the simultaneous modelling approach in terms of the number of received orders that could be considered.

As it is seen for all three cases, the sequential modelling approach could result in a higher volume of picked orders within a shift, and has a much shorter solution time. It should be noted that the difference between the number of picked orders in the two approaches mainly stemmed from the optimization gap, not the available working time for pickers. Additionally, it is evident that the total workload deviation from the average workload for all the pickers is much larger when workload equity is not considered. However, this difference becomes less noticeable as the number of orders increases. This is because with larger order volumes, pickers are working almost full-time, causing the difference between their workloads to decrease.

Fig. 4 shows the working time of each picker for each case scenario listed in Table 5. Solid red bars, which represent the results obtained from the sequential modelling approach considering workload equity, show less working time for pickers in all three cases while more orders are picked. The issue of workload inequality becomes most apparent in scenarios where sequential modelling approach is employed without workload equity considerations.

Fig. 5 illustrates the utilized man-hours as a proportion of the total available man-hours (calculated by multiplying the number of

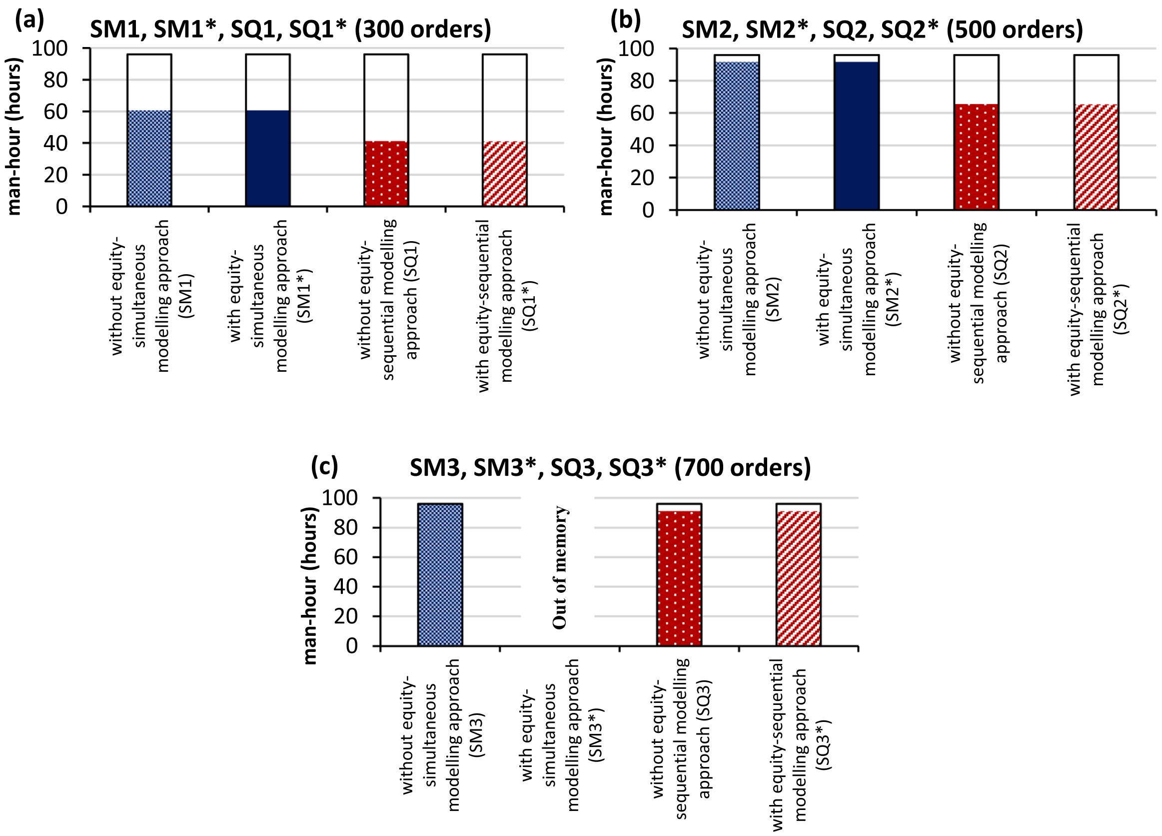

Fig. 5. (a-c). Comparison between the utilized man-hour to fulfill orders and the available man-hour at the warehouse.

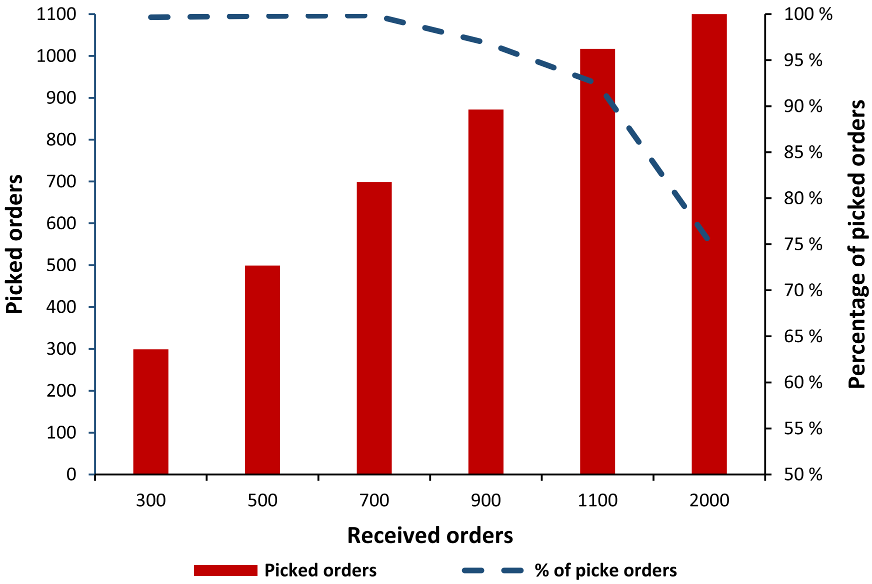

Fig. 6. Quantity and the percentage of picked orders considering different number of received orders at the warehouse.

pickers by the hours in a working shift) with the aim of maximizing the fulfilled orders based on the scenarios outlined in Table 5. It is shown that in both modelling approaches, the utilization of man-hours remains relatively consistent with and without workload balancing considerations. This consistency was anticipated, given that the number of fulfilled orders remains approximately the same with or without workload balancing considerations.

Moreover, both Figs. 4 and 5 demonstrate that under the sequential modelling approach, fewer man-hours are utilized compared to that in the simultaneous modelling approach. The reason is that in instances where the number of received orders is considerably below the warehouse capacity, the simultaneous modelling approach, which focused on maximizing the fulfilled orders, may not necessarily prioritize picking routes with minimum picking time. Conversely, the sequential modelling approach minimizes picking times for each order before maximizing the number of fulfilled orders, resulting in a reduced need for man-hours. However, this distinction becomes less noticeable when a higher number of orders are received (specifically, 700 orders and above). In such cases, to fulfill more orders, the simultaneous modelling approach must also seek picking routes with minimized picking time, diminishing the disparity between the results of the two models.

4.2. Number of picked orders for different case scenarios

The model for the sequential modelling approach is executed for cases where 300, 500, 700, 900, 1100, and 2000 orders are available to be picked. This analysis was carried out using the sequential modelling approach as this approach proved to be faster and more efficient than the simultaneous modelling approach, and could yield results for 700 orders and above.

Fig. 6 shows that with the current resources at the warehouse (i.e., pickers and picking equipment pieces), 99.7% of 300 orders, 99.8% of 500 orders, and 99.9% of 700 orders that are received at the warehouse can be picked. However, the percentage of fulfilled orders decreases to 96.9% 92.5%, and 75.3% when the number of orders placed by customers increases to 900, 1100, and 2000 orders, respectively. In the case of 2000 orders, four pickers are working for 7.9 hours, while the remaining eight pickers are working full time. The sequential modelling approach took approximately 40 min to achieve this result.

4.3. Impact of number of pickers on fulfilled orders

As expected, to pick more orders, more pickers are required. Fig. 7 shows the increase in the number of picked orders when one picker is added to the current 12 pickers for a case where 1100 orders are received at the warehouse. The simultaneous modelling approach could not yield any results due to memory shortage for this volume of orders. Therefore, this analysis was performed using only the sequential modelling approach. According to the results, on average, 2% improvement in the number of picked orders can be obtained by adding one picker.

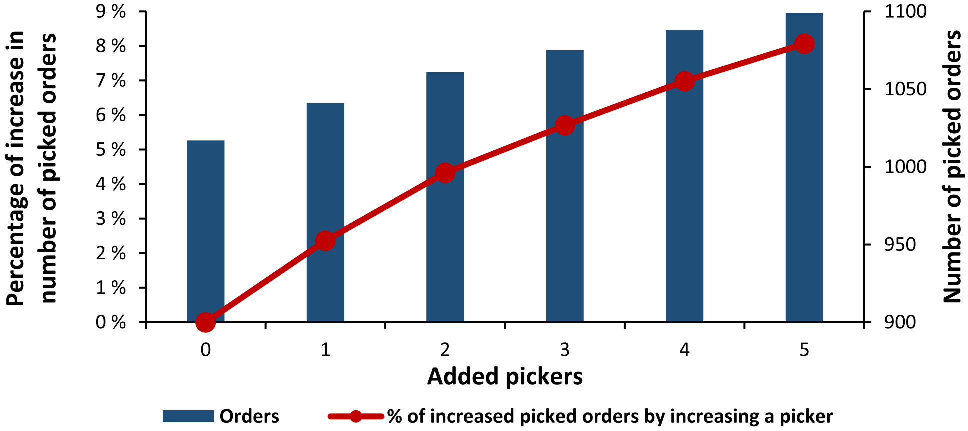

Fig. 7. Number of picked orders and percentage of improvement in the number of picked orders by adding pickers to the current number of pickers (12) working at the warehouse.

5. Discussion and managerial insights

In the majority of warehouse picking problems in the literature, the picking plan is determined by minimizing the picking time, distance, or costs, considering a certain number of orders to be picked [6,19,20]. In this way, the model would not be able to provide any insights on how to decide on the volume of orders that the warehouse can handle (i.e., fulfill). However, in this study, which is inspired by a real warehouse picking problem, instead of considering a fixed number of orders to be picked, we considered this factor as a decision variable and provided a model that maximizes the number of picked orders based on the available customer orders (i.e., demand). Maximizing the number of picked orders would indirectly affect the picking time as well. The reason is that to pick more orders, picking routes and plans with shorter picking times should be determined. However, when the number of received orders at the warehouse is considerably lower than the actual order handling capacity of the warehouse, the model would not choose the picking routes with the shortest picking time (i.e., optimized tours) necessarily by maximizing the number of picked orders (refer to Figs. 4 and 5). The reason is that all the received orders can be easily handled during the available time by the available pickers even without choosing the shortest picking tours. Therefore, only when the number of received orders is high, maximizing the number of picked orders can result in forming picking tours that are shorter and optimized in this sense. Because of this, we also provided another approach (i.e., the sequential modelling approach) for the introduced problem that maximizes the number of picked orders while determining the optimized picking tours (in terms of picking time). In this approach, the shortest picking tours for orders are determined first, and then, orders are assigned to pickers to maximize the number of picked orders. By utilizing this approach, even during periods of low order volume (i.e., customer demand), the sequence of items that pickers need to pick (i.e., pick tours) is determined in a way that pickers' working time is minimized. This way, the pickers' time and energy can be saved, allowing them to focus on other tasks in the warehouse. In general, the comparison between the two approaches shows the superiority of the sequential modelling approach. By breaking down the main problem into two stages, the complexity of the model is reduced, resulting in better outcomes in a shorter solution time.

From a practical standpoint, the proposed mathematical models for both approaches can assist warehouse managers in setting realistic expectations for the maximum number of orders that the warehouse can process within a working shift. This ultimately leads to increased customer satisfaction by ensuring timely deliveries and preventing over-promising. Additionally, by having information on the maximum order-handling capacity of the warehouse, warehouse managers can generate more revenue by avoiding the underutilization of resources.

To fulfill orders above the warehouse's maximum order-handling capacity, it is necessary to have additional picking staff. By increasing the number of pickers in the proposed model, information regarding the potential improvement in the fulfillment of orders can be provided to warehouse managers. This information enables the warehouse managers to conduct further analyses to determine if the benefits from a higher number of fulfilled orders would offset the costs of hiring new staff and acquiring additional picking equipment.

Another derived insight is that workload equity considerations make more sense for situations where a lower number of orders are received at the warehouse. As reported in Figs. 4 and 5, a higher volume of orders leads to a less noticeable difference between the working time of pickers with and without workload equity considerations. The reason is that all the pickers are working for almost the entire available time. However, when the number of available orders is lower, workload equity consideration is important, and it helps to keep pickers' schedules similar and fair. Therefore, the proposed model can help managers determine the maximum number of orders the warehouse can handle and, if the actual orders received from customers are close to this number, they do not need to worry about balancing the workload.

6. Conclusions

This research was inspired by the real case study of a warehouse of a consumer goods distribution company located in North America, which operates on a picker-to-part picking system and a strict picking policy. Two mathematical programming approaches were developed to maximize the number of picked orders during a shift in a warehouse, while taking the workload equity among the pickers into account. The models were designed to determine the order picking plans within a working shift by making decisions on the assignment of orders to the available pickers, the routes or paths that should be taken to pick each order, and the working time of each picker.

The difference between the two approaches is that the simultaneous modelling approach maximizes the number of picked orders and determines the picking tours at the same time, while the sequential modelling approach first forms the optimal picking tours in terms of picking time and then maximizes the number of orders with specific picking tours that can be assigned to pickers.

The proposed mathematical models were tested on different scenarios from the case study. The sequential modelling approach outperformed the simultaneous modelling approach in terms of the solution time, the volume of picked orders, and the working time of pickers. The performed analyses demonstrated that the order handling capacity of the warehouse decreased when more than 700 orders were received. Therefore, the warehouse should increase the number of pickers if it plans to accept and fulfill more than 700 orders per shift. An approximate 2% improvement in the number of fulfilled orders can be achieved by adding a picker.

In conclusion, the main benefit of the proposed approaches to optimize the warehouse picking is to increase the number of orders that can be picked with the same warehouse staff and assign similar workloads to pickers. This will allow the company to grow its order volume more quickly without expanding its workforce while keeping pickers' workloads fair and similar.

The warehouse optimization also contributes to reducing the tribal knowledge needed to generate picklists and picked orders, as in the current situation in the considered case study, the warehouse staff must manually generate a picklist based on their own knowledge of the warehouse and decide how to best go about picking. The warehouse picking optimization alleviates this issue and eventually will provide a circumstance under which the warehouse staff can be more productive.

This study took three practical factors into account in the order picking planning, resulting in a more practical model. However, there were other practical factors that could not be included due to a lack of proper data, which limits the effectiveness of the proposed model. For example, this study treated the searching time and retrieval time as equal for all pickers and items, while searching time of pickers is affected by pickers' experience and spatial memory. The retrieval time for different items can also be affected by the size and weight of items. Accounting for these and other practical factors [9], such as pickers skills, picker fatigue, safety constraints, and pickers' learning effect (e.g., [43,44]), can be interesting directions for future research.

The proposed models for the case study could could be solved within a reasonable timeframe. For instance, in a scenario involving 2000 received orders, 1506 orders were fulfilled with nearly all pickers operating full time. This solution was achieved within approximately 40 minutes using the sequential modelling approach. While the proposed models proved suitable for even larger problem sizes than than the size of problem in this study, future research could explore heuristic or metaheuristic algorithms to expedite solutions for large size problems.

Declaration of generative AI

During the preparation of this work, generative artificial intelligence (AI) and AI-assisted technologies were not used.

CRediT authorship contribution statement

Kimiya Rahmani Mokarrari: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Taraneh Sowlati:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jeffrey English:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Michael Starkey:** Writing – original draft, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors confirm that the majority of data supporting the findings of this study, including records of number of daily orders received at the warehouse, type and weight of the picking equipment, and number of pickers are available within the article. However, data regarding the name, specific location, and layout of the warehouse, as well as their product names and characteristics cannot be shared as it is against the internal policies of the warehouse company.

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References

[1] M.R. Islam, S.M. Ali, A.M. Fathollahi-Fard, G. Kabir, A novel particle swarm optimization-based grey model for the prediction of warehouse performance, J. Comput. Des. Eng. 8 (2) (2021) 705–727, https://doi.org/10.1093/jcde/qwab009.
[2] R. de Koster, T. Le-Duc, K.J. Roodbergen, Design and control of warehouse order picking: a literature review, Eur. J. Oper. Res. 182 (2) (2007) 481–501, https://doi.org/10.1016/j.ejor.2006.07.009.
[3] M. Masae, C.H. Glock, E.H. Grosse, Order picker routing in warehouses: a systematic literature review, Int. J. Prod. Econ. 224 (2020) 107564, https://doi.org/10.1016/j.ijpe.2019.107564.
[4] A.R.F. Pinto, M.S. Nagano, A comprehensive review of batching problems in low-level picker-to-parts systems with order due dates: main gaps, trade-offs, and prospects for future research, J. Manuf. Syst. 65 (2022) 1–18, https://doi.org/10.1016/j.jmsy.2022.08.006.
[5] Ç. Cergibozan, A.S. Tasan, Order batching operations: an overview of classification, solution techniques, and future research, J. Intell. Manuf. 30 (1) (2019) 335–349, https://doi.org/10.1007/s10845-016-1248-4.
[6] M. Farhadi Sartangi, A. Husseinzadeh Kashan, H. Haleh, A. Kazemi, A mixed integer linear formulation and a grouping league championship algorithm for a multiperiod-multitrip order picking system with product replenishment to minimize total tardiness, Complexity (2022) 1–24, https://doi.org/10.1155/2022/1382558.
[7] M. Yu, Enhancing warehouse performance by efficient order picking (2008). https://repub.eur.nl/pub/13691/EPS2008139LIS9058921673YU.pdf (accessed 11 July 2023).
[8] F. Yener, H.R. Yazgan, Optimal warehouse design: literature review and case study application, Comput. Ind. Eng. 129 (2019) 1–13, https://doi.org/10.1016/j.cie.2019.01.006.
[9] S. Vanheusden, T. van Gils, K. Ramaekers, T. Cornelissens, A. Caris, Practical factors in order picking planning: state-of-the-art classification and review, Int. J. Prod. Res. (2022) 1–25, https://doi.org/10.1080/00207543.2022.2053223.
[10] S. Vanheusden, T. van Gils, A. Caris, K. Ramaekers, K. Braekers, Operational workload balancing in manual order picking, Comput. Ind. Eng. 141 (2020) 106269, https://doi.org/10.1016/j.cie.2020.106269.
[11] S. Vanheusden, T. van Gils, K. Braekers, K. Ramaekers, A. Caris, Analysing the effectiveness of workload balancing measures in order picking operations, Int. J. Prod. Res. 60 (7) (2022) 2126–2150, https://doi.org/10.1080/00207543.2021.1884307.
[12] T. van Gils, A. Caris, K. Ramaekers, K. Braekers, R.B.M. de Koster, Designing efficient order picking systems: the effect of real-life features on the relationship among planning problems, Transp. Res. Part E Logist. Transp. Rev. 125 (2019) 47–73, https://doi.org/10.1016/j.tre.2019.02.010.
[13] Ö. Öztürkoğlu, D. Hoşer, An evaluation of order-picking tour efficiency in two-block warehouses, OSCM (2019) 74–87, https://doi.org/10.31387/oscm0370225.
[14] J.J. Bartholdi, S.T. Hackman, Warehouse & distribution science, the supply chain and logistics institute (2019). https://www.warehouse-science.com/book/editions/wh-sci-0.98.1.pdf (accessed 17 June 2024).
[15] N. Shetty, B. Sah, S.H. Chung, Route optimization for warehouse order picking operations via vehicle routing and simulation, SN Appl. Sci. 2 (2) (2020) 311, https://doi.org/10.1007/s42452-020-2076-x.
[16] M. Haouassi, Y. Kergosien, J.E. Mendoza, L.M. Rousseau, The integrated orderline batching, batch scheduling, and picker routing problem with multiple pickers: the benefits of splitting customer orders, Flex Serv. Manuf. J. 34 (3) (2022) 614–645, https://doi.org/10.1007/s10696-021-09425-8.
[17] A. Scholz, D. Schubert, G. Wäscher, Order picking with multiple pickers and due dates – simultaneous solution of order batching, batch assignment and sequencing, and picker routing problems, Eur. J. Oper. Res. 263 (2) (2017) 461–478, https://doi.org/10.1016/j.ejor.2017.04.038.
[18] T. van Gils, A. Caris, K. Ramaekers, K. Braekers, Formulating and solving the integrated batching, routing, and picker scheduling problem in a real-life spare parts warehouse, Eur. J. Oper. Res. 277 (3) (2019) 814–830, https://doi.org/10.1016/j.ejor.2019.03.012.
[19] E. Ardjmand, H. Shakeri, M. Singh, O. Sanei Bajgiran, Minimizing order picking makespan with multiple pickers in a wave picking warehouse, Int. J. Prod. Econ. 206 (2018) 169–183, https://doi.org/10.1016/j.ijpe.2018.10.001.
[20] S.A.B. Rasmi, Y. Wang, H. Charkhgard, Wave order picking under the mixed-shelves storage strategy: a solution method and advantages, Comput. Oper. Res. 137 (2022) 105556, https://doi.org/10.1016/j.cor.2021.105556.
[21] S. Saylam, M. Çelik, H. Süral, The min–max order picking problem in synchronised dynamic zone-picking systems, Int. J. Prod. Res. (2022) 1–19, https://doi.org/10.1080/00207543.2022.2058433.
[22] J.C.H. Pan, P.H. Shih, M.H. Wu, J.H. Lin, A storage assignment heuristic method based on genetic algorithm for a pick-and-pass warehousing system, Comput. Ind. Eng. 81 (2015) 1–13, https://doi.org/10.1016/j.cie.2014.12.010.
[23] X. Wang, J. Zhang, H. Shang, A real-time Synchronized Zone order picking system for balancing picker's workload, ICIC Express Lett. 4 (5) (2013) 8.
[24] C.G. Petersen, G. Aase, A comparison of picking, storage, and routing policies in manual order picking, Int. J. Prod. Econ. 92 (1) (2004) 11–19, https://doi.org/10.1016/j.ijpe.2003.09.006.
[25] R. Davidson, What is Order Picking? And What Are The Best Methods?, SoftwareConnect. https://softwareconnect.com/warehouse-management/order-picking-methods/(accessed 06 July 2023).
[26] A. Atashi Khoei, H. Süral, M.K. Tural, Energy minimizing order picker forklift routing problem, Eur. J. Oper. Res. 307 (2) (2022) 604–626, https://doi.org/10.1016/j.ejor.2022.08.038.
[27] F. Weidinger, Picker routing in rectangular mixed shelves warehouses, Comput. Oper. Res. 95 (2018) 139–150, https://doi.org/10.1016/j.cor.2018.03.012.
[28] F. Weidinger, N. Boysen, M. Schneider, Picker routing in the mixed-shelves warehouses of e-commerce retailers, Eur. J. Oper. Res. 274 (2) (2019) 501–515, https://doi.org/10.1016/j.ejor.2018.10.021.
[29] X. Xu, C. Ren, A novel storage location assignment in multi-pickers picker-to-parts systems integrating scattered storage, demand correlation, and routing adjustment, Comput. Ind. Eng. 172 (2022) 108618, https://doi.org/10.1016/j.cie.2022.108618.
[30] J.A. Cano, Formulations for joint order picking problems in low-level picker-to-part systems, Bull. EEI 9 (2) (2020) 834–842, https://doi.org/10.11591/eei.v9i2.2110.
[31] J.A. Cano, A.A. Correa-Espinal, R.A. Gómez-Montoya, Mathematical programming modeling for joint order batching, sequencing and picker routing problems in manual order picking systems, J. King Saud Univ. Eng. Sci. 32 (3) (2020) 219–228, https://doi.org/10.1016/j.jksues.2019.02.004.
[32] S. Henn, Order batching and sequencing for the minimization of the total tardiness in picker-to-part warehouses, Flex Serv. Manuf. J. 27 (1) (2015) 86–114, https://doi.org/10.1007/s10696-012-9164-1.
[33] B. Aerts, T. Cornelissens, K. Sörensen, The joint order batching and picker routing problem: modelled and solved as a clustered vehicle routing problem, Comput. Oper. Res. 129 (2021) 105168, https://doi.org/10.1016/j.cor.2020.105168.
[34] M. Yousefi Nejad Attari, A. Ebadi Torkayesh, B. Malmir, E. Neyshabouri Jami, Robust possibilistic programming for joint order batching and picker routing problem in warehouse management, Int. J. Prod. Res. 59 (14) (2021) 4434–4452, https://doi.org/10.1080/00207543.2020.1766712.

[35] T.L. Chen, C.Y. Cheng, Y.Y. Chen, L.K. Chan, An efficient hybrid algorithm for integrated order batching, sequencing and routing problem, Int. J. Prod. Econ. 159 (2015) 158–167, https://doi.org/10.1016/j.ijpe.2014.09.029.
[36] P. Yang, Z. Zhao, H. Guo, Order batch picking optimization under different storage scenarios for e-commerce warehouses, Transp. Res. Part E Logist. Transp. Rev. 136 (2020) 101897, https://doi.org/10.1016/j.tre.2020.101897.
[37] J. Liang, Z. Wu, C. Zhu, Z.H. Zhang, An estimation distribution algorithm for wave-picking warehouse management, J. Intell. Manuf. 33 (4) (2022) 929–942, https://doi.org/10.1007/s10845-020-01688-6.
[38] T. Chabot, L.C. Coelho, J. Renaud, J.F. Côté, Mathematical model, heuristics and exact method for order picking in narrow aisles, J. Oper. Res. Soc. 69 (8) (2018) 1242–1253, https://doi.org/10.1080/01605682.2017.1390532.
[39] F. Chen, H. Wang, Y. Xie, C. Qi, An ACO-based online routing method for multiple order pickers with congestion consideration in warehouse, J. Intell. Manuf. 27 (2) (2016) 389–408, https://doi.org/10.1007/s10845-014-0871-1.
[40] A. Silva, L.C. Coelho, M. Darvish, J. Renaud, Integrating storage location and order picking problems in warehouse planning, Transp. Res. Part E Logist. Transp. Rev. 140 (2020) 102003, https://doi.org/10.1016/j.tre.2020.102003.
[41] S. Ghotb, T. Sowlati, J. Mortyn, D. Roeser, V.C. Griess, A goal programming model for the optimization of log logistics considering sorting decisions and social objective, Can. J. For. Res. 52 (5) (2022) 716–726, https://doi.org/10.1139/cjfr-2021-0203.
[42] C.E. Miller, A.W. Tucker, R.A. Zemlin, Integer programming formulation of traveling salesman problems, J. ACM 7 (4) (1960) 326–329, https://doi.org/10.1145/321043.321046.
[43] J. Zhang, X. Zhang, N. Zhang, Considering pickers' learning effects in selecting between batch picking and batch-synchronized zone picking for online-to-offline groceries, Appl. Math. Model. 113 (2023) 358–375, https://doi.org/10.1016/j.apm.2022.09.009.
[44] E.H. Grosse, C.H. Glock, An experimental investigation of learning effects in order picking systems, J. Manuf. Technol. Manage. 24 (6) (2013) 850–872, https://doi.org/10.1108/JMTM-03-2012-0036.